Change of Sign Method

Exam code: 9MA0

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Written by: Lucy Kirkham

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Change of sign method

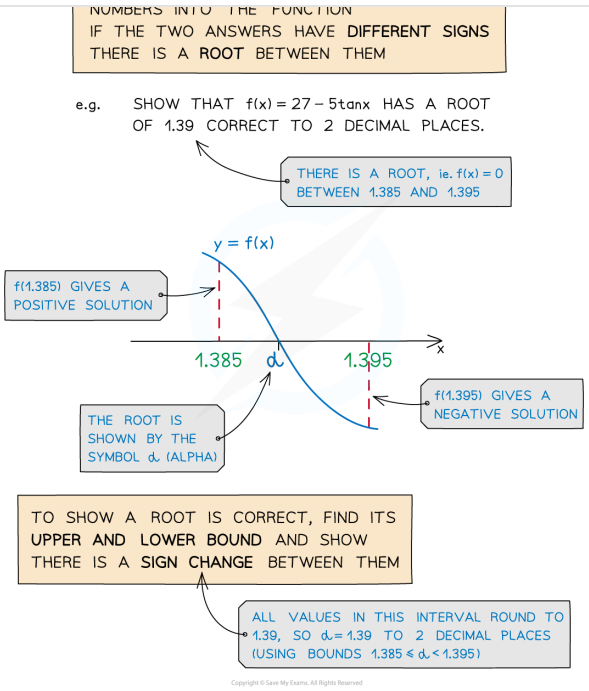
What does a change of sign mean?

- A sign change between f(a) to f(b) means there must be a root between a and b
- When an equation cannot be solved using the usual **analytical methods**, we can use change of sign to find **approximate solutions**
- We can say a root lies within a particular interval or is correct to a given accuracy
- Bounds can be used to show a root is correct
- Using sign change to find a root is only appropriate for continuous functions in a small interval

Stuck on something?

Ask questions and get help with exactly what you're revising.

Got it



Examiner Tips and Tricks

- Remember this will only work if the function is continuous and the interval is small enough.
- Sign change questions may be part of bigger numerical methods questions.

Worked Example

the x-axis at the point P (p, 0).

a) Show that $0.7 < p < 0.8$.

b) Show that $p = 0.771$ to 3 decimal places.

$f(x) = 3xe^x - 5$

a) $f(0.7) = 3 \times 0.7e^{0.7} - 5 = -0.771119...$ ← NEGATIVE

SUBSTITUTE VALUES INTO f(x)

$f(0.8) = 3 \times 0.8e^{0.8} - 5 = 0.34129...$ ← POSITIVE

CHANGE OF SIGN MEANS $0.7 < p < 0.8$

b) $p = 0.771$ (3dp) ← FIND UPPER AND LOWER BOUND OF p

$0.7705 \leq p < 0.7715$

SUBSTITUTE BOUNDS INTO f(x)

$f(0.7705) = -0.0052...$ ← NEGATIVE

$f(0.7715) = 0.0062...$ ← POSITIVE

SOLUTION MUST BE $p = 0.771$ TO 3dp

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